L3.pdf
PDF

You are an advanced academic scribe. Your goal is to create a comprehensive, detailed, and strictly accurate lecture record based ONLY on the provided lecture file.

Strict Constraints:

Source Material: Use ONLY the text, equations, and images found in the attached document. Do not hallucinate or bring in outside assumptions.

Detail Level: High. Do not summarize. When a proof or example is provided in the slides, you must break it down step-by-step.

Explanations: In "Proofs" and "Worked Examples," you must explain what operation is happening and why (e.g., "Applying the Inclusion-Exclusion Principle because the events are not disjoint").

Final Output: After generating the full text in the chat, you must run Python code to generate a downloadable PDF of this scribe.

Structure (Follow the format and depth of this example exactly):

[ONE-SHOT EXAMPLE START]

Lecture Scribe: Fundamentals of Geometry

Lecture 1: Quadrilateral Properties

Name: [Student Name] | Enrollment No: [Student ID]

Table of Contents

Properties of Rectangles

Topic 1: Properties of Rectangles

1. Definitions and Notation

Definition 1.1 (Rectangle): A quadrilateral is defined as a rectangle if it has four interior right angles (90°). It is denoted as $\square ABCD$.

Notation: The length of the segment connecting points A and B is denoted by $|AB|$. The area is denoted by $\mathcal{A}$.

2. Theorems and Results

Theorem 1.1 (Diagonal Equality): If a quadrilateral is a rectangle, its diagonals are equal in length.

Mathematical Statement: If $\square ABCD$ is a rectangle, then $|AC| = |BD|$.

3. Proofs and Derivations

Proof of Theorem 1.1:

Objective: We aim to prove $|AC| = |BD|$ using triangle congruence.

Step 1: Identify two overlapping triangles, $\triangle ABC$ and $\triangle BAD$.

Step 2: Observe that $|AD| = |BC|$.

Reasoning: Opposite sides of a rectangle are equal by definition.

Step 3: Observe that $\angle DAB = \angle CBA = 90^\circ$.

Reasoning: All angles in a rectangle are right angles (Definition 1.1).

Step 4: Observe that side $|AB|$ is common to both

triangles.

Reasoning: Reflexive property.

Step 5: Apply the SAS (Side-Angle-Side) Congruence Criterion.

Conclusion: $\triangle ABC \cong \triangle BAD$.

Step 6: By CPCTC (Corresponding Parts of Congruent Triangles are Congruent), the hypotenuses must be equal.

Final Result: $|AC| = |BD|$. Q.E.D.

4. Worked Examples

Example 1.1: Calculation of Diagonal Length

Problem Statement: Consider a rectangle with width $w=3$ and height $h=4$. Calculate the length of the diagonal d.

Solution Steps:

Identify the relationship: A rectangle contains right angles. Therefore, the diagonal divides the rectangle into two right-angled triangles. We can use the Pythagorean Theorem.

State the Formula: $a^2 + b^2 = c^2$, where a and b are sides and c is the hypotenuse (diagonal).

Substitution: Substitute $w=3$ and $h=4$ into the equation:

$$d^2 = 3^2 + 4^2$$

Calculation:

$$d^2 = 9 + 16 = 25$$

Final Step: Take the square root to find length.

$$d = \sqrt{25} = 5$$

Answer: The diagonal length is 5 units.

[ONE-SHOT EXAMPLE END]

Task:

Using the structure and depth of the example above, generate the Lecture Scribe for the attached lecture file.

You must write the full content in Markdown.

Below is the complete, strictly source-based lecture scribe generated **only** from the provided lecture file L3 .

Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor: Dr. Dhaval Patel, PhD
Date: January 13, 2026

Table of Contents

1. Course Introduction
2. Motivational Foundations
3. Growth Mindset vs Fixed Mindset
4. CSE400: General Course Information
5. Course Website and Active Learning Platform
6. Schedule and Lecture Sessions
7. Discussion & Communication Guidelines
8. Important Notes
9. Project Component (30%)
10. Project Execution Milestones (M1–M6)

11. Submission Guidelines
12. Undergraduate Research Programme (UGRP)
13. MICxN Research Lab Overview
14. Research Outcomes and Awards
15. Current Students
16. Closing Remarks

---

1. Course Introduction

1.1 Course Title

CSE400 – Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Instructor details (as displayed on title slide):

- Dr. Dhaval Patel, PhD
- Associate Professor
- Major Advisor – Computer Science and Engineering (CSE)
- Machine Intelligence, Computing and xG Networks (MICxN) Research Lab
- SEAS – Ahmedabad University, Ahmedabad, Gujarat – India

---

2. Motivational Foundations

2.1 Quote Slide

Slide content includes the following quotation:

> "Good is the enemy of great. That's why so few things become great."
> — James C. Collins

This slide emphasizes striving beyond mediocrity toward excellence.

2.2 Growth Mindset

The slide titled **"Growth Mindset"** includes the following statements:

- "Failure is an opportunity to grow"
- "I like to try new things"

- “I can learn to do anything I want”
- “Challenges help me grow”
- “My effort and attitude determine my abilities”
- “Feedback is constructive”
- “I am inspired by the success of others”

The word **GROWTH mindset** is prominently displayed.

2.3 Growth vs Fixed Mindset

The next slide contrasts:

Growth Mindset

(Statements listed above)

Fixed Mindset

- “Failure is the limit of my abilities”
- “I’m either good at it or I’m not”
- “My abilities are unchanging”
- “I don’t like to be challenged”
- “My potential is predetermined”
- “When I’m frustrated, I give up”
- “I stick to what I know”

This comparison establishes the mindset framework expected in the course.

---

3. Course Outline

3.1 Outline – Part 1

CSE400: General Course Information

- Team
- Active Learning Platform: Campuswire
- Schedule
- Grading

3.2 Why Should We Learn CSE400?

- Example: Daily life conversations

3.3 Engineering Applications

- Speech Recognition
- System Radar System
- Communication Network

---

4. CSE400: General Course Information

4.1 Team

Instructor

Dr. Dhaval Patel

- Role: Instructor
- Place: Faculty Office (Room-210)
- Faculty Profile: https://ahduni.edu.in/seas/people/faculty/dhaval-patel ↗
- Personal Webpage: http://profpatel.in/ ↗
- Area of Interest:
  - xG Networks
  - Applied ML
  - DL / RL
  - AutoML
  - Intelligent Transportation Systems
  - Life Sciences
  - Behaviour Modelling using AI
- Email: dhaval.patel@ahduni.edu.in ↗

4.2 Teaching Assistants / Team Members

Deep Patel

- BTech-CSE (3rd Year)
- Working in RL and Pinching Antenna Systems
- Email: deep.p4@ahduni.edu.in ↗

Prapti Patel

- BTech-CSE (4th Year)
- Smart Sensing Framework Using Kolmogorov-Arnold Networks for Enhanced Detection in 5G Environments (COMSNETS'25)
- FAN-F: A Fourier Analysis Network-based framework for efficient and reliable Spectrum Sensing for 5G and Beyond (CCNC'26)
- Email: prapti.p@ahduni.edu.in ↗

Raj Koticha

- BTech-CSE (4th Year)
- Proactive Self-Aware Multi-Agent Reinforcement Learning for Resource Management in NR-V2X Platooning (COMSNETS'26)
- Email: raj.k1@ahduni.edu.in ↗

Ritu Patel

- BTech-CSE (4th Year)
- Working in Intelligent Transportation System
- Email: rituben.p@ahduni.edu.in ↗

Rushi Moliya

- BTech-CSE (4th Year)
- Optimizing UAV Deployment for Enhanced Detection Performance in Multi-UAV Cooperative Sensing (NCC 2025)
- Optimizing Multi-UAV 3D Deployment for Energy-Efficient Sensing over Uneven Terrains (arXiv 2025)
- Email: rushi.m@ahduni.edu.in ↗

Ura Modi

- BTech-CSE (3rd Year)
- Working in Pinching Antenna Systems
- Email: ura.m@ahduni.edu.in ↗

---

5. Active Learning and Class Discussion

5.1 Course Website

Section-1: https://campuswire.com/c/G20541104/ ↗

Section-2: https://campuswire.com/c/G766840CE/ ↗

5.2 Why Use Campuswire?

1. To build confidence level:
   - Participate anonymously
   - Post / Back-channel communication during lecture
2. Collaborative and Active Learning:
   - Solve difficulties with online participation
3. Feedback on lecture session:
   - Real-time feedback
   - Polling
4. Personal Communication with Instructor/TAs:
   - DM – Direct Message communication

---

6. Schedule

6.1 Lecture Sessions

Section-1:

- 9:30 AM to 11:00 AM
- Tuesday, Thursday
- GICT Room -136

Section-2:

- 1:00 PM to 2:30 PM
- Tuesday, Thursday
- GICT Room -137

6.2 TA Hours

- In person / Online: TAs
- Timings: Yet to be finalized – A post will be made soon.

---

7. Connect with the Instructor

Timing for Discussion and Difficulty Sessions

- Contact Hours: 24 × 7 through Campuswire

- Best Practice: Put your queries on Campuswire
- Need more time: Feel free to DM – Private conversation
- External Engagement:
  - UGRP-8: 2026
  - Join with Offline Projects
  - Need Counseling
  - Cup of TEA for 10 Min: Write on office email ID

---

8. Important Note

- LaTeX Tutorial and Assignment Submission Guideline

---

9. CSE400 – Project Kickoff (30%)

9.1 Project Team Formation

Deadline:

17th January 2026, Saturday – EOD

---

10. Project Execution Guidelines

Weightage

- 30%

Major Milestones (Total 6 Submissions – 1 per group)

1. Concept Evolution Maps
2. Scribe: Process & Decision-Making
   - Decision Logs (Why X over Y?)
   - Constraints
   - Alternatives considered
   - Final decision
   - Evidence used
   - Trade-off Matrices (Cost vs performance vs risk tables)
3. Multimodal Artifacts (Video / Audio / Visual)
   - Think-Aloud Videos
   - One-Minute Insight Videos

    - Project Demo
4. Question-Driven Artifacts
5. Collaboration & Team Dynamics Artifacts

Deliverables:

- Codes
- Reports
- Videos
- As specified

Team Assessment:

- Before and After Midsemester

Project Viva and Submission:

- Towards the end of the course

ChatGPT 5.2 Save as PDF

M1: Kickstart

- Team Formation
- Area Identification
- Background
- Motivation
- Problem Formulation

M2: Mathematical Modeling

- Mathematical Modelling of the problem in ANY domain
    - RV
    - PMF/PDF
    - CDF
    - Multivariate RVs
    - Joint PMF/PDF/CDF

M3: Coding

- Simulation + Computation

M4: Inference

- Choose, Understand, and code a randomized algorithm

M5: Randomized Algorithms

- Apply randomized algorithm to domain problem
- Compare with deterministic algorithm

M6:

- Derive Bounds
- Analysis
- Compile and submit final deliverables

---

12. Submission Details

Submission #1 – Concept Evolution Maps

Tools:

- https://miro.com/concept-map/ ↗
- https://app.diagrams.net/ ↗ (draw.io)

Submission #2 – Scribe: Learning Reflection Logs

Types:

- Lecture Scribes
- Project Scribe

Requirements:

- Every lecture: 2 groups assigned
- Reflect lecture content
- Include additional examples from foreign textbooks
- Minimum 8–10 pages

Total:

- 6 submissions
- Bi-weekly mode

Submission #3 – Multimodal Artifacts

- One video per milestone
- 10–15 minutes
- Explain milestone
- Coding simulation (if applicable)

- PPT/Google Slides allowed
- Any tool for recording
- Focus on content (not editing skills)

---

13. Undergraduate Research Programme (UGRP)

Introduction to UGRP-8 (2026–2027)

Dr. BJ Fogg:

- Founded the Behavior Design Lab at Stanford University
- Directs research and innovation

Why UGRP?

Reference:

Tranquillo, Joe.

"The t-shaped engineer."

Journal of Engineering Education Transformations 30.4 (2017): 12–24.

IBM Concept of T-Shaped Individual

- Vertical bar → Depth in single technical discipline
- Horizontal bar → Ability to apply knowledge across disciplines and work with others

---

Philosophy for UGRP

- Multidisciplinary: Arts, Science, Management
- Experiential Learning
- Research Driven
- 4D Model:
  - Discover
  - Design
  - Develop
  - Deliver
- Breadth: CS and CSE
- Data Science: Applied Artificial Intelligence

- Modern Computer System Design: Hardware and Software
- Networks and IoT / IoBNT / IoV

---

14. MICxN Research Lab

Industry Activities

Design and Development of 5G enabled Intelligent Transportation Systems Testbed in Gujarat

- Basic Communication via V2X SL between OBU and RSU
- SDK Testing
- OBU VIC interfacing and sensor data collection
- ITS in-car Dashboard Development

Research:

- Deep Learning based Crash Prevention Framework
- Deep-RL based Smart Signaling
- Position aided mmWave Beam Prediction
- C-V2X and Wi-Fi Coexistence Framework

International Bilateral Collaborations

(Listed professors and affiliations including University of Liverpool, University of Manchester, Boston College, Monash University, NTU Singapore, Qatar University, University of York, etc.)

Alumni (MICxN Research Lab)

Includes alumni working at:

- Google
- Amazon
- Intel
- American Express
- Simplify Healthcare
- Arizona State University

Published Venues

- ICSigSys
- COMSNETS'25
- NCC'25
- COMSNETS'26
- IEEE Transactions on Vehicular Technology

Awards:

- Best UG Presentation Award
- Best Paper Award

---

15. Current Students (MICxN Lab)

UG Thesis Scholars:

- Dr. Dhaval Patel (Lab Director)
- Prapti Patel – Advanced Spectrum Sensing
- Raj Koticha – O-RAN
- Rushi Moliya – 6G ISAC + UAV
- Ritu Patel – 6G ITS Testbed

Other Students:

- Siya Dhaduk – Wi-Fi Sensing
- Kaushika Rathod – Wireless Networks
- Charmi Goyal – Wi-Fi 8
- Hiya Shah – Wi-Fi 8
- Ura Modi – Pinching Antennas
- Devashree Parikh – O-RAN
- Deep Patel – Pinching Antennas
- Shayan Pariyal – Pinching Antennas

---

16. Closing

Q & A Session: Open Discussions

Thank you!

Email: dhaval.patel@ahduni.edu.in ↗

PDF Generation

Now generating the downloadable PDF file of this lecture scribe.

Sources